

```

%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;
scuts_path = 'e:\SHARE\Fe\Data\analysis\07_fit_with_multicut_allJ\sel_cuts';
Cutfitfile = fullfile(scuts_path, 'NG_FFCutsFits.mat');

if ~exist('NG_FFCutsFits', 'var')
    ld = load(Cutfitfile);
    NG_FFCutsFits = ld.fit_src_struct;
end

```

```

fn = fieldnames(NG_FFCutsFits);
nf = 11

```

```

nf =
11

```

```

%for i = 11:numel(fn)
%selectCuts = fn{nf};
selectCuts = 'Ei400_off200NGmaxFF'

```

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selectCuts =
'Ei400_off200NGmaxFF'

```

```

if contains(selectCuts, 'msk')
    error('Can not fit field: %s', selectCuts);
end
if contains(selectCuts, 'all_fit_par')
    error('Can not fit field: %s', selectCuts);
end

fprintf('Processing cuts: %s \n', selectCuts);

```

```

Processing cuts: Ei400_off200NGmaxFF

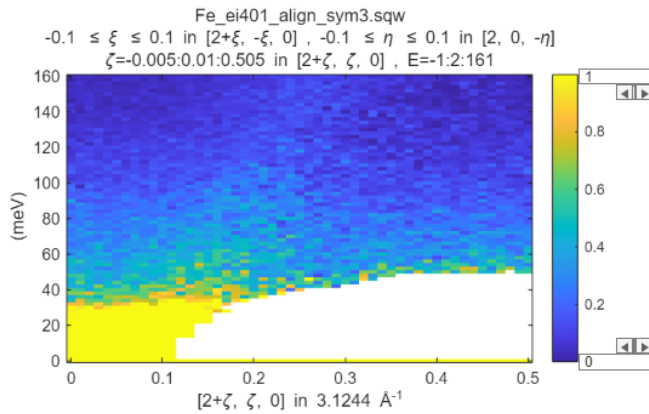
```

```

cuts_list = NG_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts, 'Ei200')
    mi = maps_instrument(200, 600, 'S');
end
if contains(selectCuts, 'Ei400')
    mi = maps_instrument(401, 600, 'S');
end
if contains(selectCuts, 'Ei800')
    mi = maps_instrument(786, 600, 'S');
end
sample=IX_sample(true, [1,0,0], [0,1,0], 'cuboid', [0.04,0.03,0.02]);
for j=1:numel(cuts_list)
    cuts_list{j} = cuts_list{j}.set_instrument(mi);
end

```

```
cuts_list{j} = cuts_list{j}.set_sample(sample);
plot(cuts_list{j}); lz 0 1; keep_figure;
end
```



```
dir_name = "NG";

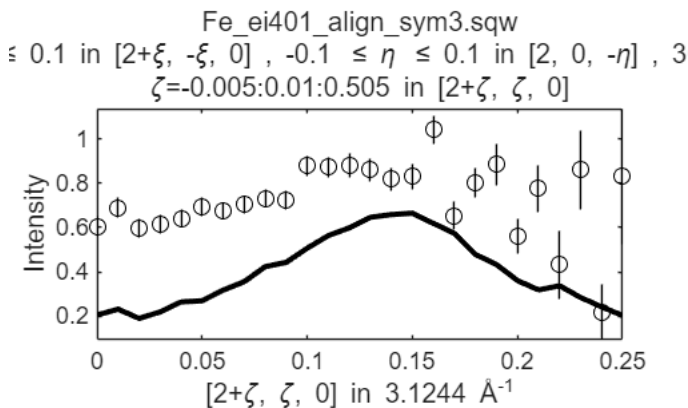
dE_step = 2; %original energy transfer step data were binned to. No point in going
finer
half_dE = 5; % half width of data binning

en_mm = cuts_list{1}.data.img_range(:,4);
en_mm(1) = max(30,en_mm(1));
%en_mm(1) = 100;
%en_mm(2) = 380;
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);

cut_dimensions = 1;
NG_FFCutsFits.(selectCuts) = cuts_list;

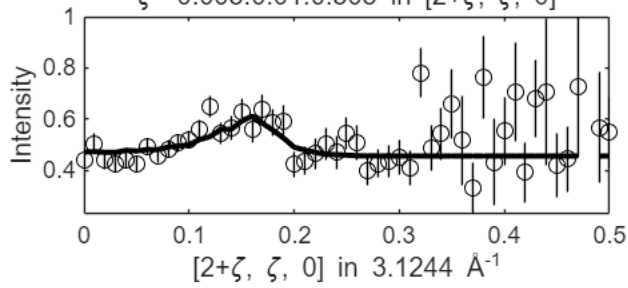
NG_FFCutsFits = fit_multicutsFF_along_direction(...
    NG_FFCutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
```

```
*****
**** En = 35±5 going to 155
*****
```

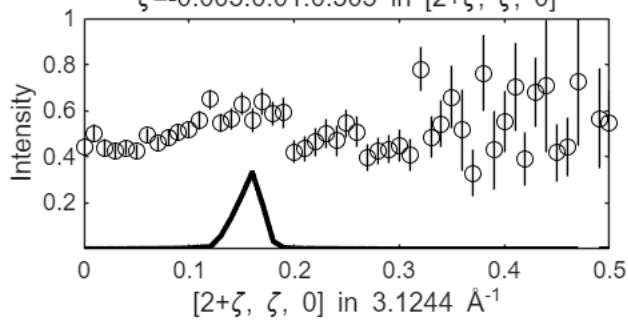


```
*****
**** En = 45±5 going to 155
*****
```

Fe_ei401_align_sym3.sqw
J0= 20± 1.6; J1= 0± 0; gamma=-0.946248
 $\xi \in [2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 4
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$

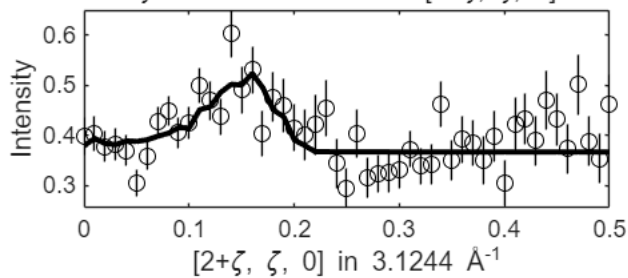


Fe_ei401_align_sym3.sqw
 $\xi \in [2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 4
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$

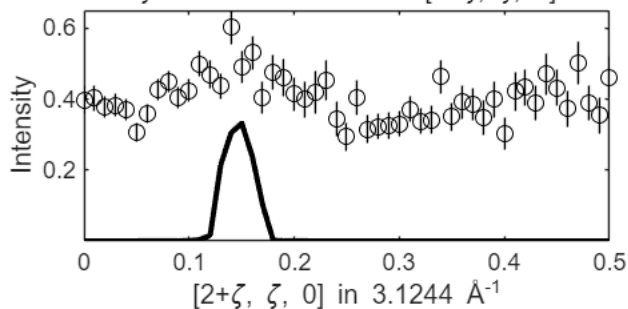


**** En = 55±5 going to 155

Fe_ei401_align_sym3.sqw
J0= 27±0.26; J1= 0± 0; gamma=-0.188919:
 $\xi \in [2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 5
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



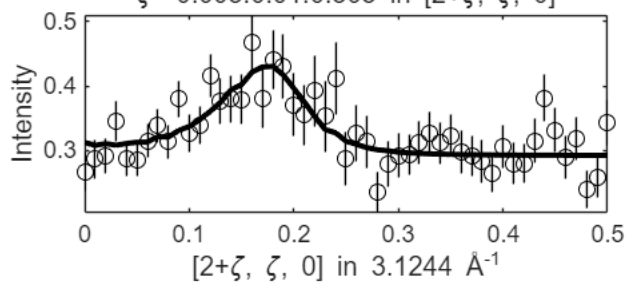
Fe_ei401_align_sym3.sqw
 $\xi \in [2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 5
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



**** En = 65±5 going to 155

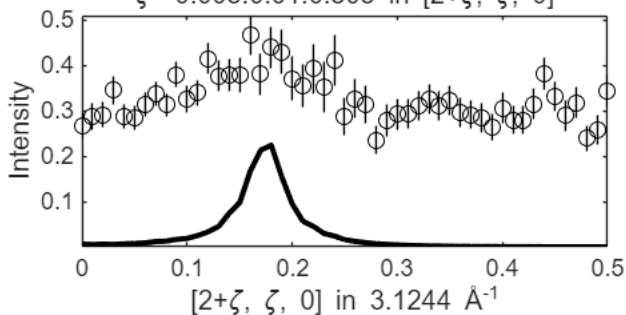
Fe_ei401_align_sym3.sqw

1 ± 0.15 ; $J_0 = 25 \pm 1.4$; $J_1 = 0 \pm 0$; $\gamma = 10.6527$
 $\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 6
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



Fe_ei401_align_sym3.sqw

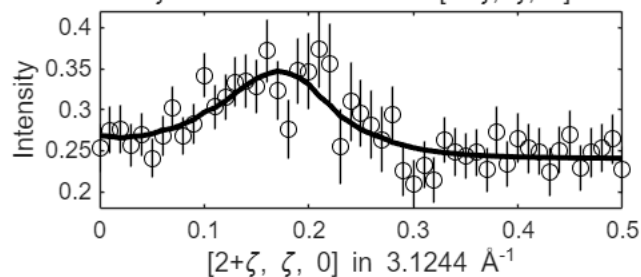
$\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 6
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



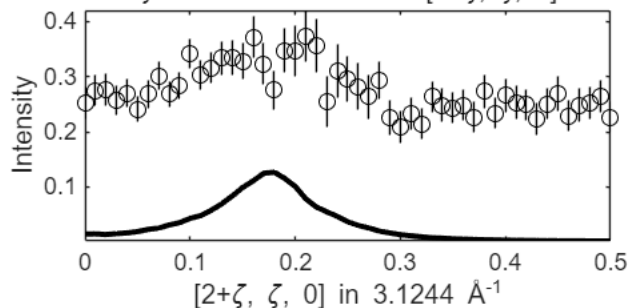
**** En = 75±5 going to 155

Fe_ei401_align_sym3.sqw

1 ± 0.34 ; $J_0 = 30 \pm 2.7$; $J_1 = 0 \pm 0$; $\gamma = 37.4958$
 $\xi \leq 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 7
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$

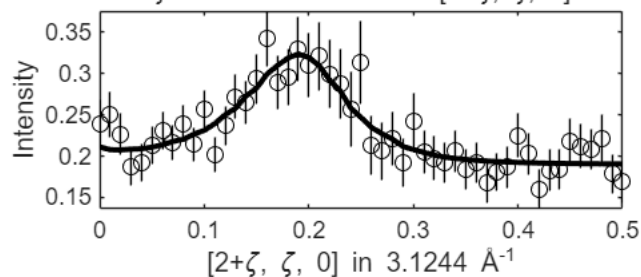


Fe_ei401_align_sym3.sqw
 $\xi \in [2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 7
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$

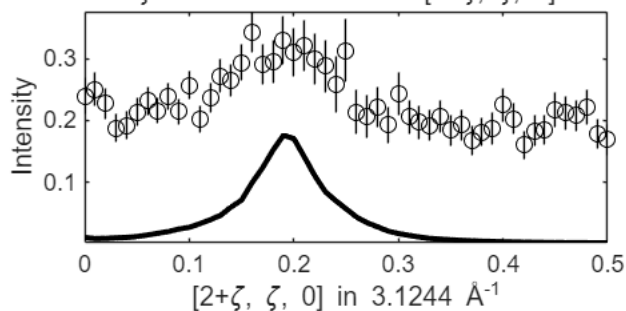


**** En = 85±5 going to 155

Fe_ei401_align_sym3.sqw
 1 ± 0.19 ; $J_0 = 28 \pm 1.1$; $J_1 = 0 \pm 0$; $\gamma = 25.3656$;
 $\xi \in [2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 8
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$

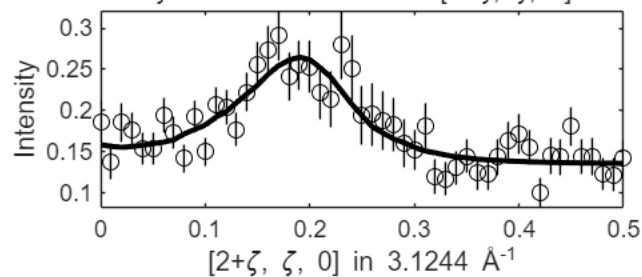


Fe_ei401_align_sym3.sqw
 $\xi \in [2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 8
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$

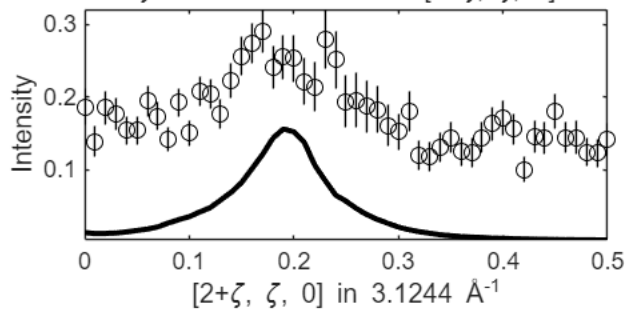


**** En = 95±5 going to 155

Fe_ei401_align_sym3.sqw
 ± 0.097 ; $J_0 = 33 \pm 1.7$; $J_1 = 0 \pm 0$; $\gamma = 35.9628$
 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 90°
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$

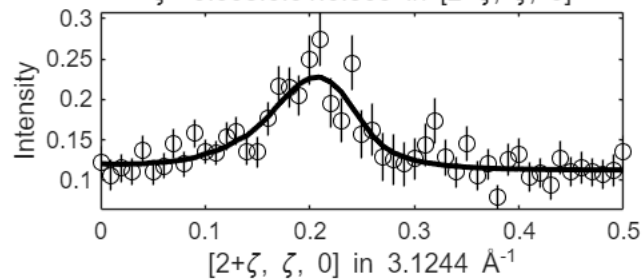


Fe_ei401_align_sym3.sqw
 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 90°
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$

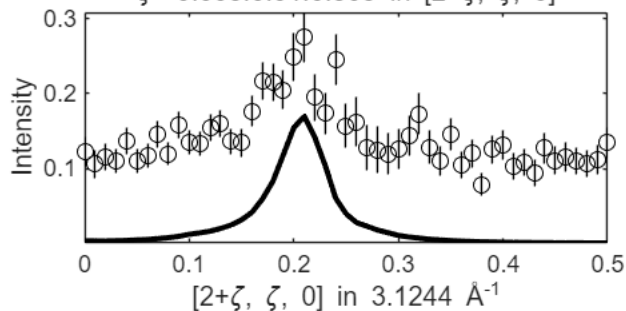


 **** $E_n = 105 \pm 5$ going to 155

Fe_ei401_align_sym3.sqw
 ± 0.14 ; $J_0 = 31 \pm 1.1$; $J_1 = 0 \pm 0$; $\gamma = 18.5767$
 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 10°
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



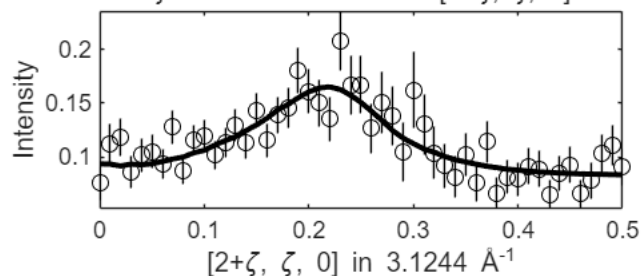
Fe_ei401_align_sym3.sqw
 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 10°
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



**** En = 115±5 going to 155

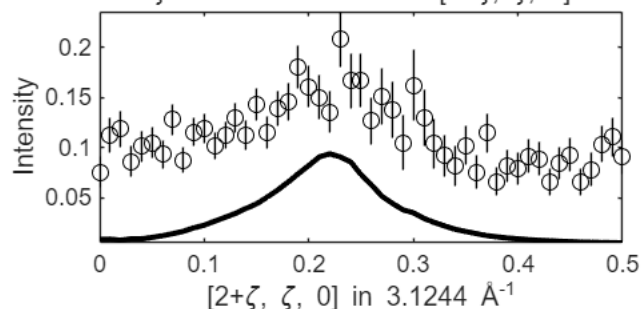
Fe_ei401_align_sym3.sqw

$\lambda \pm 0.45$; $J_0 = 34 \pm 2.5$; $J_1 = 0 \pm 0$; $\gamma = 57.6959 \pm 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 11
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



Fe_ei401_align_sym3.sqw

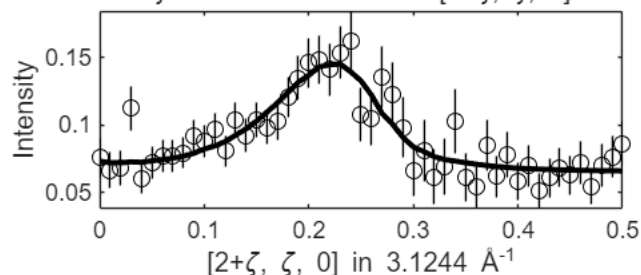
$\lambda \pm 0.45$; $J_0 = 34 \pm 2.5$; $J_1 = 0 \pm 0$; $\gamma = 57.6959 \pm 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 11
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



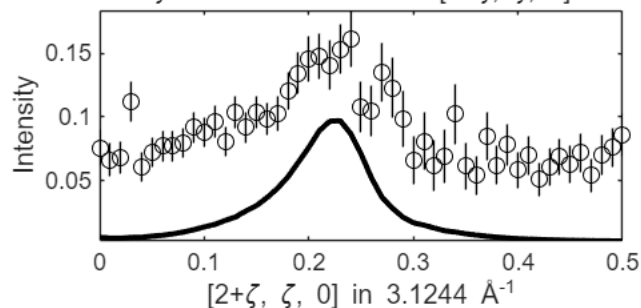
**** En = 125±5 going to 155

Fe_ei401_align_sym3.sqw

$\lambda \pm 0.17$; $J_0 = 35 \pm 1.2$; $J_1 = 0 \pm 0$; $\gamma = 36.9764 \pm 0.1$ in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 12
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$

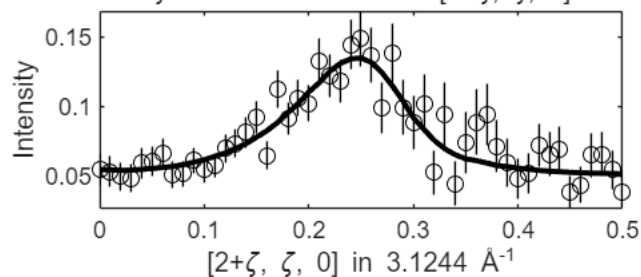


Fe_ei401_align_sym3.sqw
0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 12
 $\zeta=-0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$

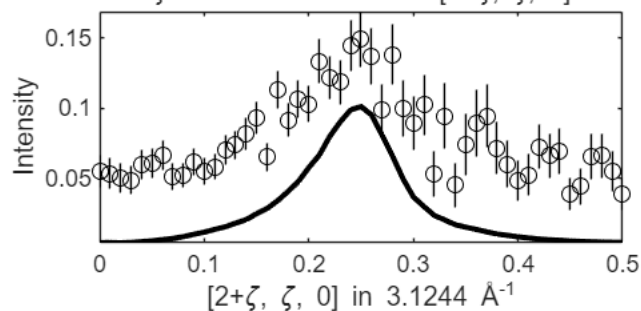


**** En = 135 ± 5 going to 155

Fe_ei401_align_sym3.sqw
 $\beta \pm 0.19$; $J_0 = 33 \pm 0.96$; $J_1 = 0 \pm 0$; $\gamma = 39.8797 \pm$
0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 13
 $\zeta=-0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$

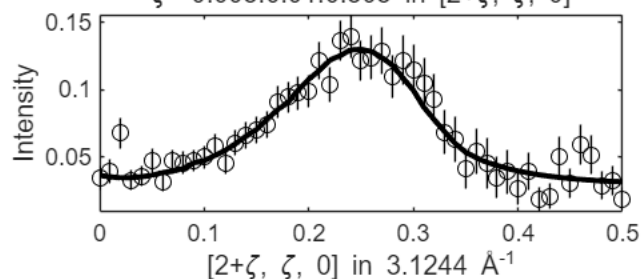


Fe_ei401_align_sym3.sqw
0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 13
 $\zeta=-0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$

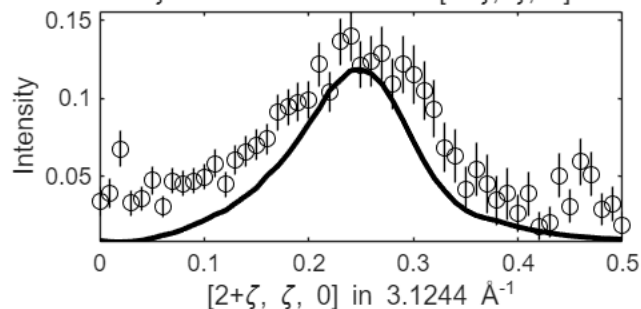


**** En = 145 ± 5 going to 155

Fe_ei401_align_sym3.sqw
 $\lambda \pm 0.15$; $J_0 = 36 \pm 0.72$; $J_1 = 0 \pm 0$; $\gamma = 64.6611$;
 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 14
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$

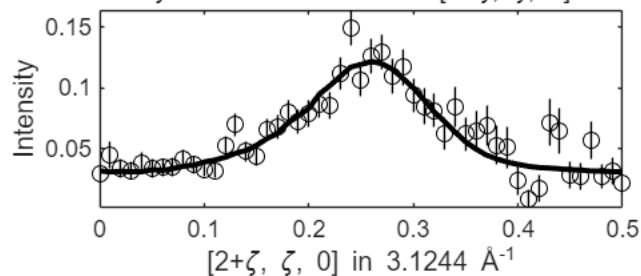


Fe_ei401_align_sym3.sqw
 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 14
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$

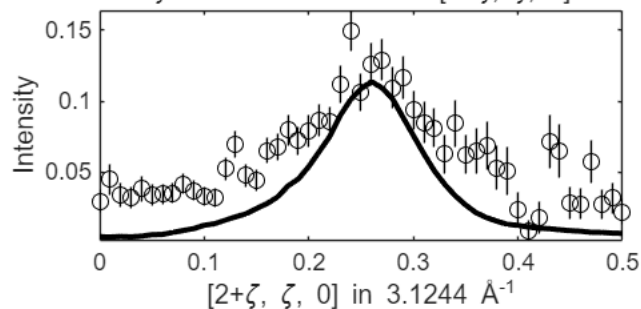


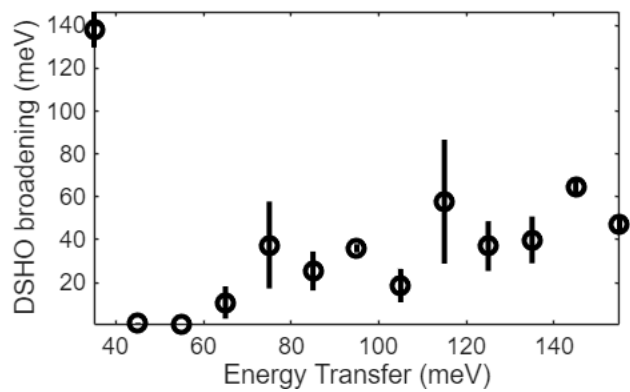
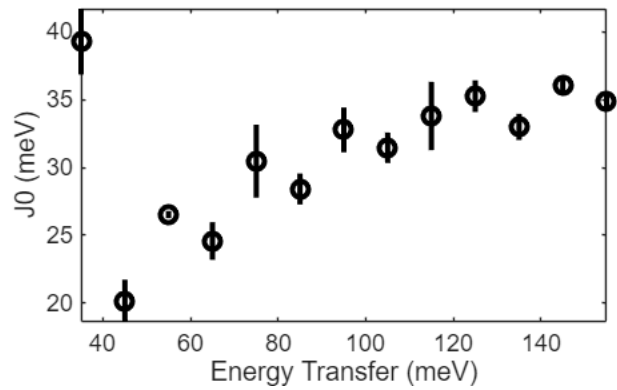
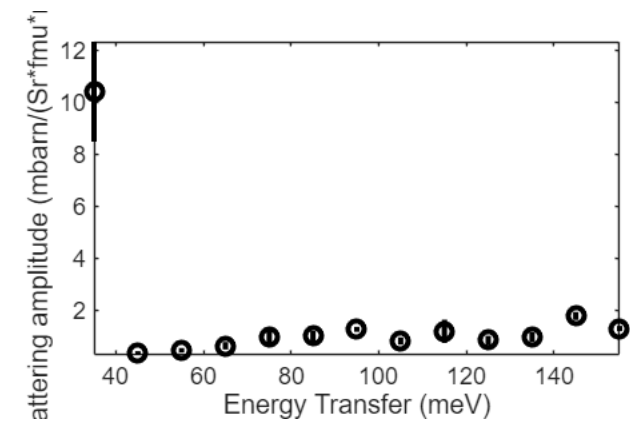
**** $E_n = 155 \pm 5$ going to 155

Fe_ei401_align_sym3.sqw
 $\lambda \pm 0.1$; $J_0 = 35 \pm 0.73$; $J_1 = 0 \pm 0$; $\gamma = 47.0505$;
 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 15
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$



Fe_ei401_align_sym3.sqw
 0.1 in $[2+\xi, -\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[2, 0, -\eta]$, 15
 $\zeta = -0.005:0.01:0.505$ in $[2+\zeta, \zeta, 0]$





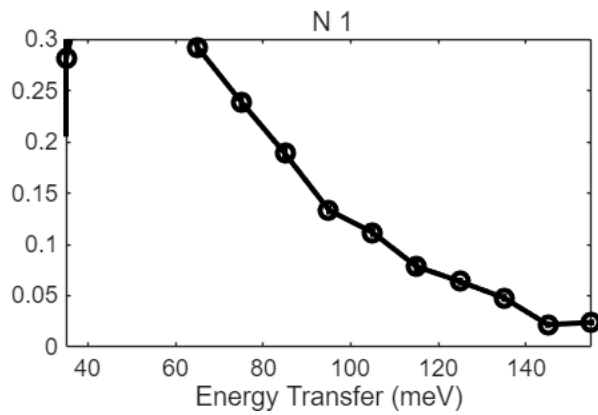
```

c1 = NG_FFCutsFits.(selectCuts);
NG_FFCutsFits.(selectCuts) = [c1{:}];

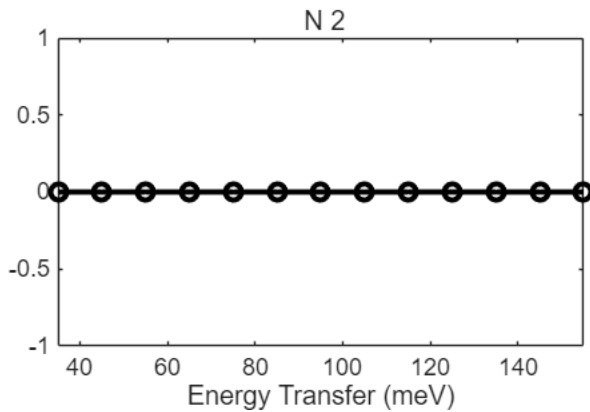
beep
beep
beep

all_fit_par = NG_FFCutsFits.([selectCuts, '_all_fit_par']);
bg_data = extract_bg_par(all_fit_par, [1, 2]);
% original slope parameters
pd(bg_data(1)); lx(all_fit_par(1).en, all_fit_par(end).en); ly 0 0.3; keep_figure;

```



```
pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure
```



```
%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_figure
```

```
doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_NG.mlx';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_NG_',selectCuts,'.pdf']);
export(thisFile,targ_file);
%close all;
%end
```

```
fit_src_struct = NG_FFCutsFits;
save(Cutfitfile,"fit_src_struct",-v7.3')
```